

Requirement Analysis

Solution Requirements (Functional & Non-functional)

Date	29-06-2026
Project Name	Automated Network Request Management
Team ID	
Maximum Marks	

Functional Requirements

These are the specific behaviors, features, and functions your ServiceNow system must perform based on your workflow design.

FR no.	Requirement Description	Success / Acceptance Criteria
FR-1	Input Validation in Request Form	ServiceNow catalog form should validate fields like IP addresses using correct IPv4/IPv6 format before submission is allowed.
FR-2	External System Integration	Integration Hub should convert approved request data into JSON and send it securely to network systems via REST APIs.
FR-3	Post-Deployment Verification	System must automatically check network connectivity after implementation to confirm successful changes.target network controllers via secure REST APIs.

FR-4	CMDB Synchronization	After completion, the system must update CMDB records and maintain an audit log for traceability.
FR-5	CMDB Auto-Update & Auditing	Upon successful fulfillment, the system must write updated configuration item (CI) relationships directly to the CMDB and log an immutable audit trail on the ticket record.
FR-6	Automated Notifications	The system must send automated notifications (via email or chat webhooks) to the requestor at major lifecycle stages: submission, approval, and completion.

Non-functional Requirements

These define the system's operational attributes—how well the system performs its functions. Since you are building on the ServiceNow platform, you inherit many of these automatically, but they must still be documented.

ID	Requirement Description	Success / Acceptance Criteria
NFR-1	Performance / Speed	Automated request fulfillment (from the moment of final approval to successful API execution) must complete in less than 15 minutes for standard paths.

NFR-2	Security & Authentication	All outbound API calls from ServiceNow Integration Hub to network infrastructure must use encrypted protocols (HTTPS/SSH) and credential aliases stored securely in the platform.
NFR-3	Scalability	The workflow orchestration system must be capable of processing up to 500 concurrent automated network request transactions without causing API timeouts or platform lag.
NFR-4	Availability & Reliability	The automated request engine must achieve 99.9% uptime during operational hours, with graceful error-handling that routes failed API attempts to a manual NetOps queue.
NFR-5	Usability (UX)	Intake forms must use dynamic UI policies to show/hide fields, ensuring a minimalist layout that minimizes user input confusion and reduces incomplete submissions.